Answer Key - Biology - Undergraduate Level Assessment 1 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** A
   Options: A) High detrital biomass; B) High number of predatory species; C) High presence of fully matured plant species; D) High frequency of R-selected species; E) High rates of soil nutrient depletion
   Note: Early-successional plant communities are characterized by a rapid accumulation of organic matter from dead plant material, leading to high detrital biomass as these communities establish and grow in nutrient-rich, disturbed environments.

2. **Answer:** E
   Options: A) It increases the acid concentration in the mitochondria matrix.; B) It decreases the acid concentration in the mitochondria matrix.; C) It increases the concentration of H+ in the mitochondria matrix.; D) It decreases the OH- concentration in the mitochondria matrix.; E) It increases diffusion of H+ from the intermembrane space to the matrix.
   Note: The correct answer is based on the principle that a low external pH increases the concentration gradient of hydrogen ions (H+) across the inner mitochondrial membrane, promoting the diffusion of H+ from the intermembrane space into the matrix, which drives ATP synthesis through ATP synthase.

3. **Answer:** A
   Options: A) the tRNA transcript from the original DNA; B) a single strand of the original DNA segment after a substitution mutation; C) the final processed mRNA made from the original DNA; D) the primary RNA transcript (after splicing) from the original DNA; E) a single strand of the original DNA segment after a duplication mutation
   Note: The tRNA transcript from the original DNA contains the fewest number of nucleotides because tRNA is typically much shorter than mRNA and only contains the specific sequences necessary for translating a particular amino acid, whereas mRNA includes the entire coding region along with introns and regulatory sequences.

4. **Answer:** A
   Options: A) thrombin; B) glycosaminoglycans; C) collagens; D) fibroblasts
   Note: Thrombin is a blood-clotting enzyme involved in the coagulation process and is not a structural component of connective tissue, which primarily consists of cells, fibers, and ground substance.

5. **Answer:** A
   Options: A) Water is useful just because of its cooling properties.; B) Water is only important for hydration.; C) Water's role is limited to acting as a chemical reactant in cellular processes.; D) Water is important primarily because it contributes to the structural rigidity of cells.; E) Water's only function is to provide a medium for aquatic

organisms.
Note: Water's cooling properties, due to its high specific heat capacity and ability to absorb heat without significantly changing temperature, play a crucial role in regulating the internal temperatures of living organisms and maintaining homeostasis, making it essential for life.

True / False

6. **Answer:** True
   Options: A) True; B) False
   Note: The neurilemma sheath, which is formed by Schwann cells, plays a crucial role in nerve regeneration by providing a supportive environment and a physical pathway for the axon to reconnect with its target tissue after injury.

7. **Answer:** False
   Options: A) True; B) False
   Note: Genetic engineering is a human-directed process that alters specific traits, while the main source of genetic variation among human populations is natural genetic mutations and recombination during reproduction.

8. **Answer:** True
   Options: A) True; B) False
   Note: The macronucleus in paramecium is essential for regulating metabolic processes and gene expression, and its removal disrupts these vital functions, ultimately leading to cell death.

9. **Answer:** True
   Options: A) True; B) False
   Note: The statement is true because osmosis causes water to move from an area of lower solute concentration (the surrounding pure water) to an area of higher solute concentration (inside the potato cells), resulting in the potato swelling as it absorbs water.

10. **Answer:** False
    Options: A) True; B) False
    Note: An increase in the amount of CO 2 in the blood leads to a rise in carbonic acid, which decreases blood pH and stimulates the respiratory center in the brain to increase the rate and depth of respiration to expel excess CO 2 and restore pH balance.

Long Form Response

11. **Answer:** Hydrochloric acid (HCl) plays a crucial role in the stomach by creating an acidic environment that aids in the digestion of food and helps kill harmful bacteria ingested with food. The presence of HCl activates digestive enzymes, particularly pepsin, which breaks down proteins into peptides. In the absence of HCl, as seen in

conditions like achlorhydria, the stomach becomes less effective at controlling bacterial populations, leading to an increased risk of gastrointestinal infections and dysbiosis. Additionally, without sufficient acidity, the digestion of proteins is compromised, which can hinder the absorption of essential nutrients and ultimately affect overall gut health. Thus, HCl is vital not only for digestion but also for maintaining a balanced and healthy gut microbiome.
Note: correct because it highlights hydrochloric acid's essential role in creating an acidic environment for protein digestion and bacterial control, and it identifies the negative consequences of its absence, such as impaired digestion and increased risk of infections.

12. **Answer:** The parathyroid glands, typically four small glands located on the posterior surface of the thyroid gland, play a crucial role in calcium regulation within the human body. They primarily produce parathyroid hormone (PTH), not insulin, which is responsible for increasing calcium levels in the blood by promoting the release of calcium from bones, increasing calcium absorption in the intestines, and enhancing renal reabsorption of calcium. Dysfunction of the parathyroid glands, such as in hyperparathyroidism or hypoparathyroidism, can lead to significant health issues, including osteoporosis, kidney stones, or neuromuscular irritability. Proper functioning of these glands is essential for maintaining calcium homeostasis and overall metabolic health. Thus, understanding the structure and function of the parathyroid glands is vital in the context of endocrine health and disease management.
Note: correct because it accurately describes the structure and function of the parathyroid glands, specifically their role in calcium regulation through the secretion of parathyroid hormone (PTH), and addresses the health implications of their dysfunction.

End of Answer Key